



Quantifying Biological Responses

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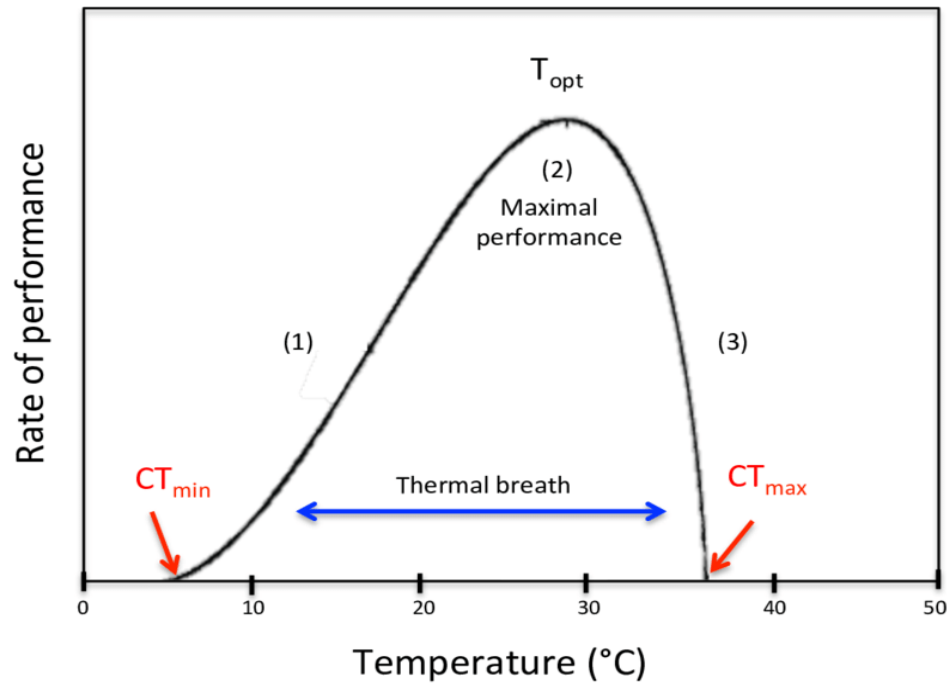
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QUANTIFYING RESPONSES AT SPECIES LEVEL

Species-Specific Responses

Thermal Tolerance

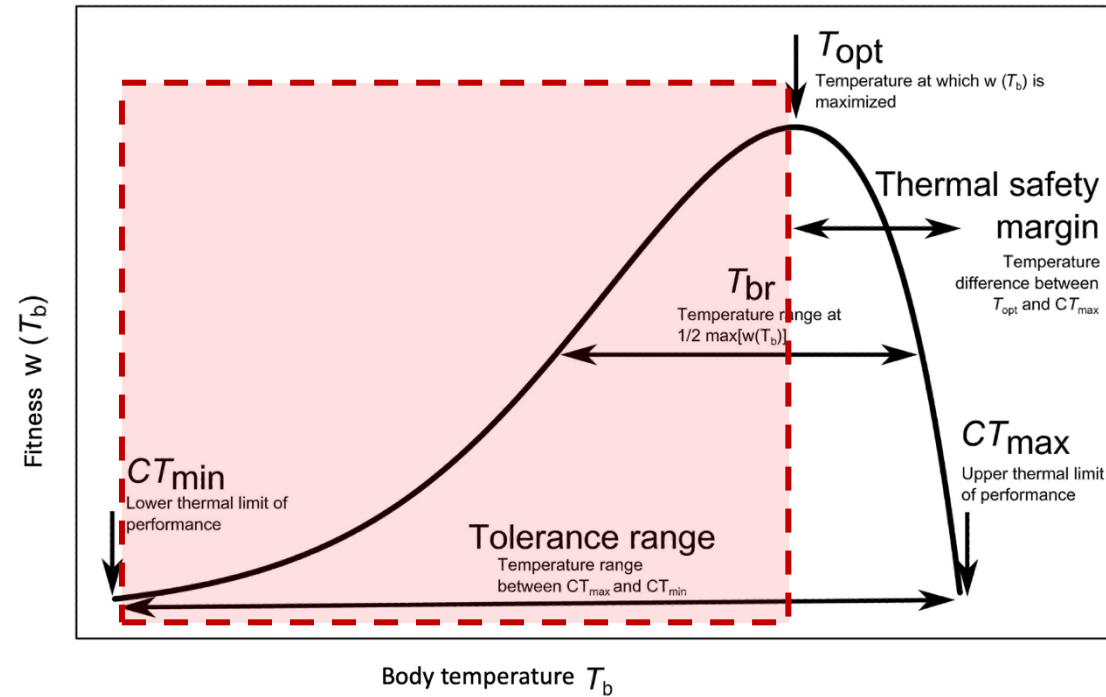


Why are TPCs asymmetrical?

- Performance/fitness falls off much faster above T_{opt} than it rises below it
- Driven by underlying biological mechanisms:

1. **Enzyme Kinetics (Exponential Rising Phase):** As temperature increases below T_{opt} , biochemical reaction rates accelerate according to standard thermodynamic principles (Q_{10} temperature coefficient). Reaction rates increase exponentially with temperature.
2. At T_{opt} , Enzyme reaches maximum activity
3. **Denaturation and Stress (falling phase):** Once temperatures exceed T_{opt} , critical enzymes begin to denature, cell membrane structure breaks down, and oxygen supply to tissues fails to meet metabolic demand, causing rapid physiological collapse.

Thermal Tolerance



Enzyme Kinetics (Exponential Rising Phase):

- Q_{10} temperature coefficient measures the rate of change of a biological, chemical or physical process as a result of increasing the temperature by 10°C .
- Quantifies the temperature dependence of reaction rates
- Q_{10} calculations are **only valid within the exponential phase** (below T_{opt}) where the thermal response follows typical kinetic principles.

$$Q_{10} = \left(\frac{R_2}{R_1} \right)^{\frac{10}{T_2 - T_1}}$$

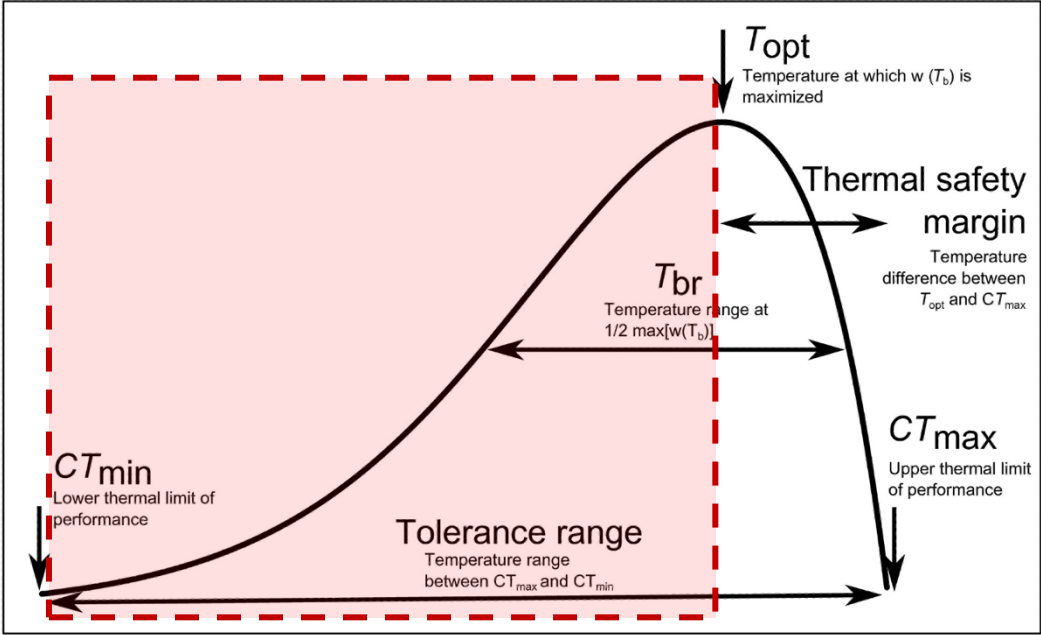
- R_1 : The measured reaction rate at initial temperature T_1 .
- R_2 : The measured reaction rate at final temperature T_2 .
- T_1 & T_2 : Temperatures in degrees Celsius ($^\circ\text{C}$) or Kelvin (K), where $T_2 > T_1$.

Thermal Tolerance

Enzyme Kinetics (Exponential Rising Phase):

$$Q_{10} = \left(\frac{R_2}{R_1}\right)^{\frac{10}{T_2 - T_1}}$$

Fitness $w(T_b)$

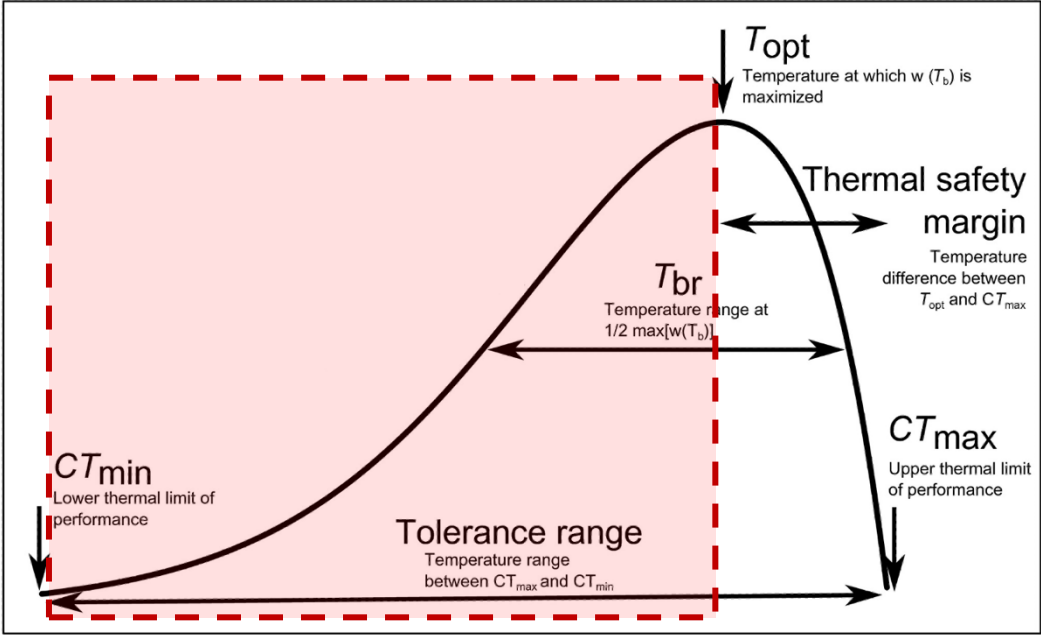


Body temperature T_b

Q_{10} Value	Typical Process Type	Example
$Q_{10} \approx 1.0$	Physical processes / Passive transport	Simple ion diffusion, photochemical reactions
$Q_{10} \approx 2.0 - 3.0$	Standard chemical & enzymatic reactions	Most enzyme-catalyzed metabolic rates (doubles/triples per 10 °C)
$Q_{10} > 3.0$	High temperature sensitivity	Ion channel gating, thermal protein denaturation

Thermal Tolerance

Fitness $w(T_b)$



Body temperature T_b

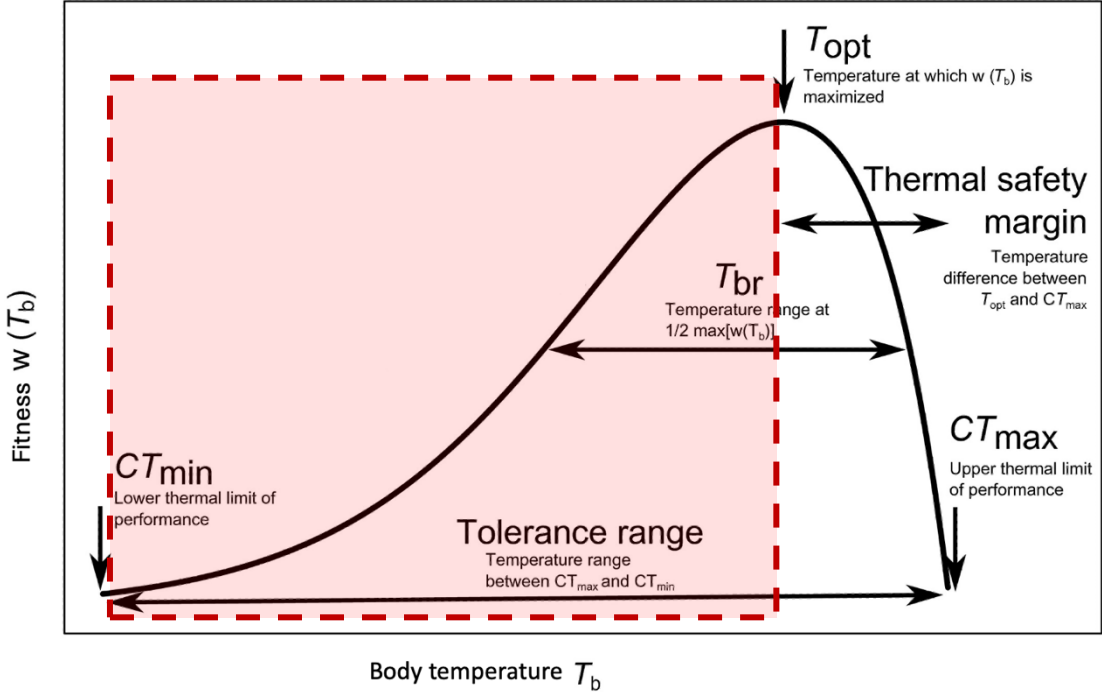
Enzyme Kinetics (Exponential Rising Phase):

$$Q_{10} = \left(\frac{R_2}{R_1} \right)^{\frac{10}{T_2 - T_1}}$$

Suppose an experiment measures the reaction velocity (V) of the enzyme *Lactate dehydrogenase* across two temperatures within its biological working range:

Parameter	Temperature 1 (T_1)	Temperature 2 (T_2)
Temperature (T)	15 °C	22 °C
Reaction Rate (R)	12.5 $\mu\text{mol/min}$	21.0 $\mu\text{mol/min}$

Thermal Tolerance



Enzyme Kinetics (Exponential Rising Phase):

$$Q_{10} = \left(\frac{R_2}{R_1} \right)^{\frac{10}{T_2 - T_1}}$$

Parameter	Temperature 1 (T_1)	Temperature 2 (T_2)
Temperature (T)	15 °C	22 °C
Reaction Rate (R)	12.5 $\mu\text{mol/min}$	21.0 $\mu\text{mol/min}$

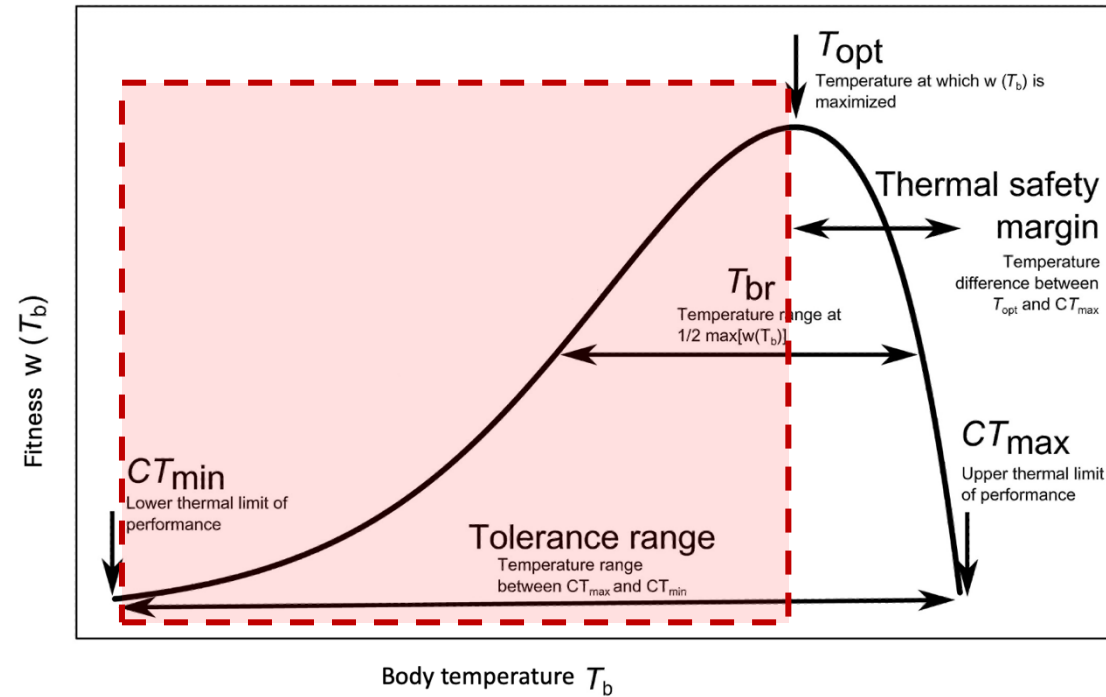
The Rate Ratio ($\frac{R_2}{R_1}$): $\frac{R_2}{R_1} = \frac{21.0}{12.5} = 1.68$

Temperature Difference Scaling Factor ($\frac{10}{T_2 - T_1}$): $\Delta T = T_2 - T_1 = 22 - 15 = 7 \text{ } ^\circ\text{C}$

Exponent = $\frac{10}{7} \approx 1.4286$

$Q_{10} = (1.68)^{1.4286} \approx 2.11$

Thermal Tolerance



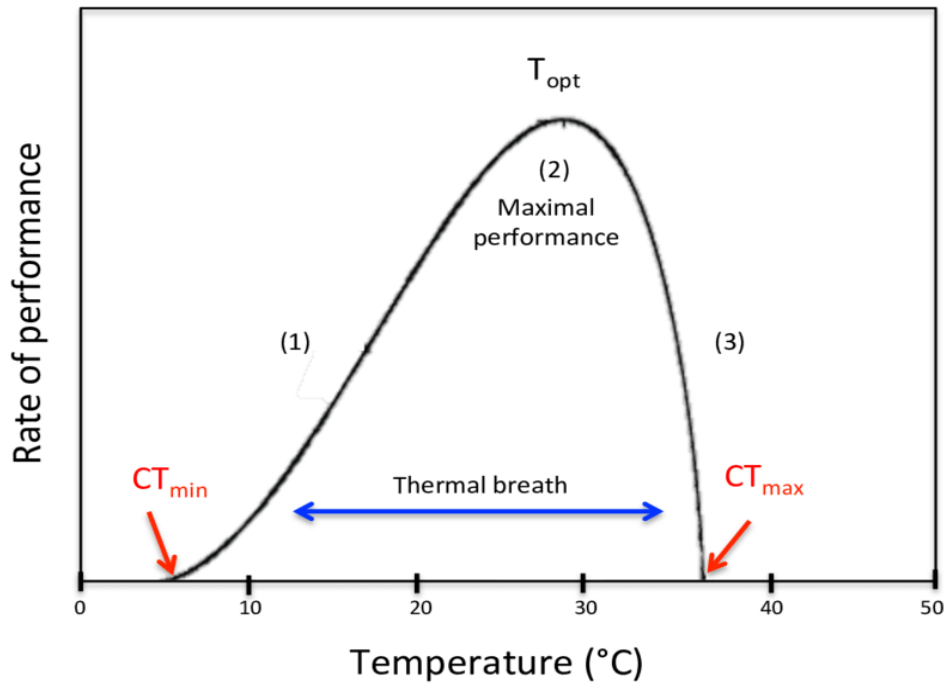
Enzyme Kinetics (Exponential Rising Phase):

$$Q_{10} = \left(\frac{R_2}{R_1} \right)^{\frac{10}{T_2 - T_1}}$$

$$Q_{10} = (1.68)^{1.4286} \approx 2.11$$

- $Q_{10} = 2.11$: This falls directly in the typical **2.0 to 3.0 range** expected for standard enzyme-catalyzed reactions.
- It means that within this temperature range, raising the temperature by a full 10°C will cause the enzymatic reaction rate to roughly **double** (specifically, multiply by 2.11).

Thermal Tolerance



Fitting Thermal Performance Curves, $P(T)$

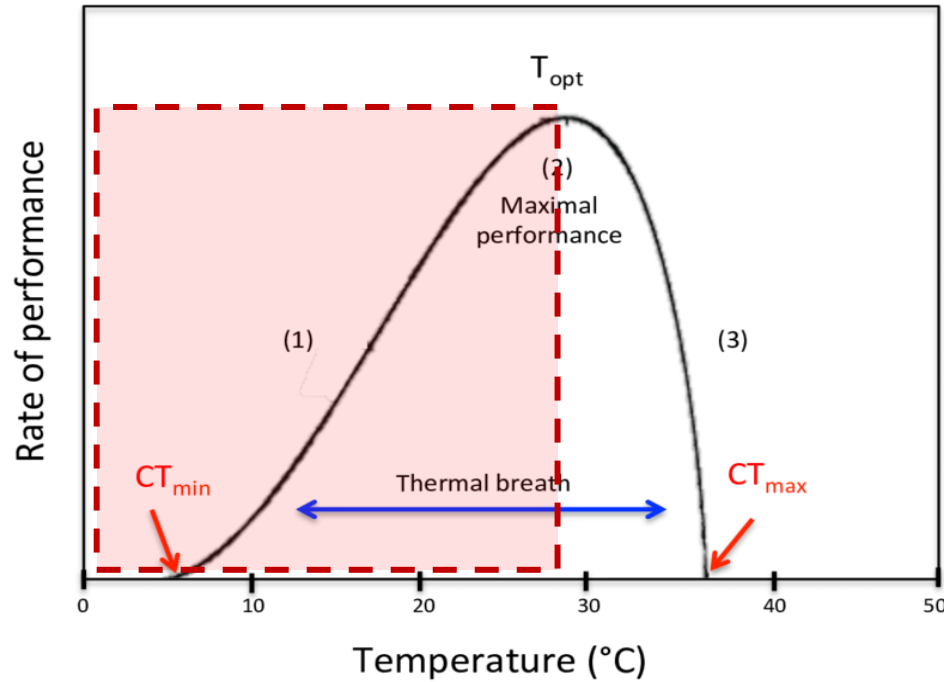
And Predicting Performance/fitness

- Use a Gaussian function to describe the rise in performance up to the optimal temperature (T_{opt})
- And a quadratic decline to zero performance at CT_{max} and higher temperatures

$$P(T) = \begin{cases} \exp\left\{-\left(\frac{T - T_{opt}}{2\sigma_p}\right)^2\right\} & \text{for } T \leq T_{opt} \\ 1 - \left(\frac{T - T_{opt}}{T_{opt} - CT_{max}}\right)^2 & \text{for } T > T_{opt} \end{cases}$$

Thermal Tolerance

Fitting Thermal Performance Curves, $P(T)$



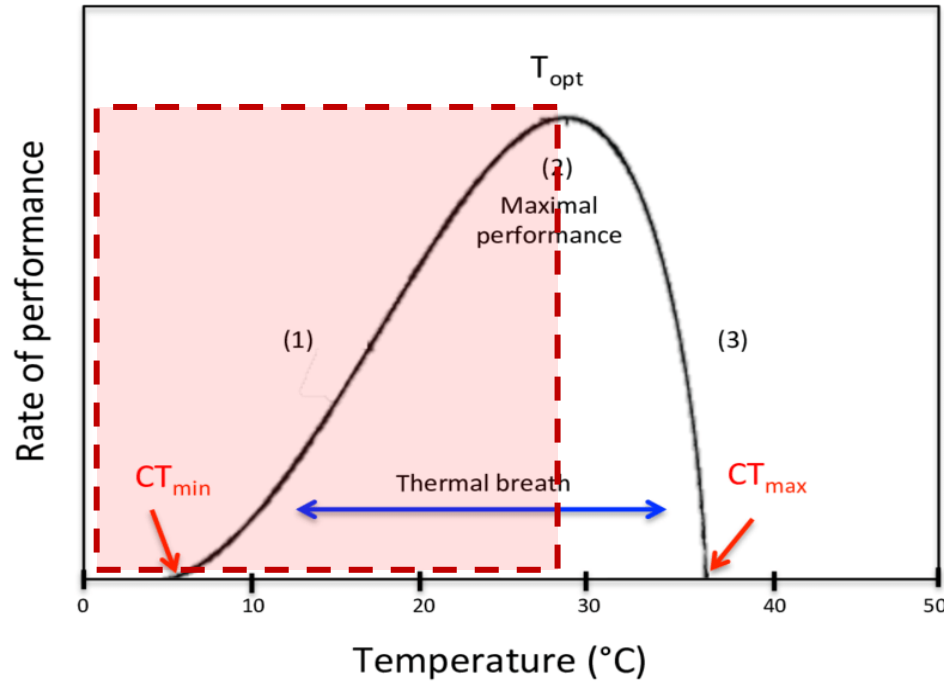
Rise in performance up to the optimal temperature (T_{opt}) ($T \leq T_{opt}$)

$$P(T) = \exp \left\{ - \left(\frac{T - T_{opt}}{2\sigma_p} \right)^2 \right\}$$

- σ_p (Performance Width): Controls the width or spread of the cold side curve.
- A larger σ_p means the organism can maintain decent performance over a broader range of cold temperatures.

Thermal Tolerance

Fitting Thermal Performance Curves, $P(T)$



Rise in performance up to the optimal temperature (T_{opt})

$(T \leq T_{opt})$

$$P(T) = \exp \left\{ - \left(\frac{T - T_{opt}}{2\sigma_p} \right)^2 \right\}$$

Rearrange to Solve for σ_p

Take the natural logarithm (ln) of both sides to remove the exp:

$$\ln(P(T)) = - \left(\frac{T - T_{opt}}{2\sigma_p} \right)^2$$

Multiply by -1 and take the square root:

$$\sqrt{-\ln(P(T))} = \frac{|T - T_{opt}|}{2\sigma_p}$$

Cross-multiply:

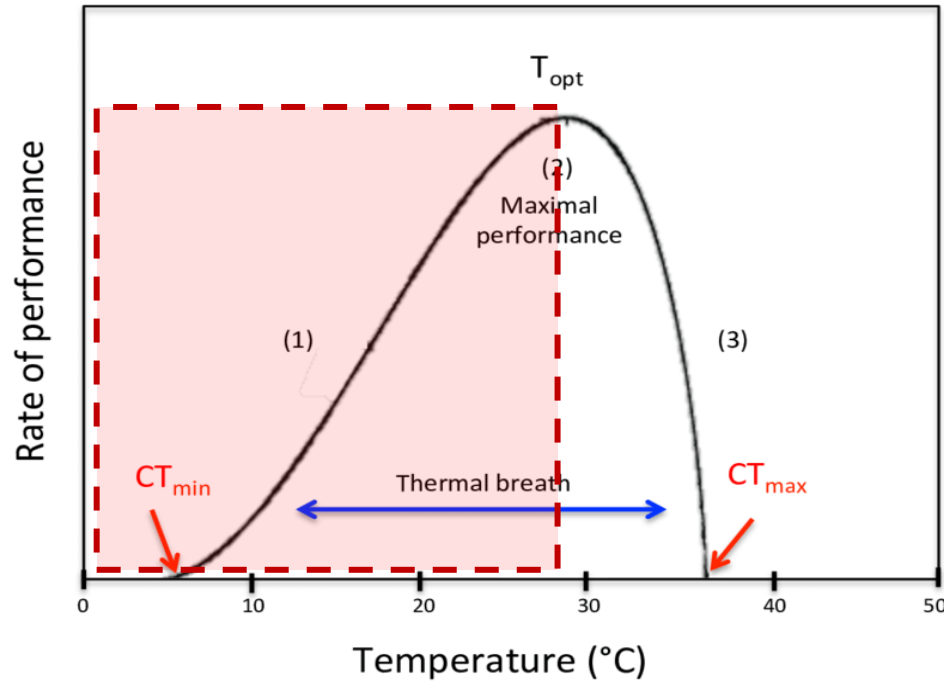
$$2\sigma_p \cdot \sqrt{-\ln(P(T))} = |T - T_{opt}| \cdot 1$$

Divide both sides by $2\sqrt{-\ln(P(T))}$ to isolate σ_p :

$$\sigma_p = \frac{|T - T_{opt}|}{2\sqrt{-\ln(P(T))}}$$

Thermal Tolerance

Fitting Thermal Performance Curves, $P(T)$



Rearrange to Solve for σ_p

$$\sigma_p = \frac{|T - T_{opt}|}{2\sqrt{-\ln(P(T))}}$$

Using the data point at $T = 10^\circ\text{C}$ for Species A where performance $P(10) = 10\% = 0.10$, and $T_{opt} = 28^\circ\text{C}$:

Temperature Difference:

$$|T - T_{opt}| = |10 - 28| = 18$$

Logarithmic Factor:

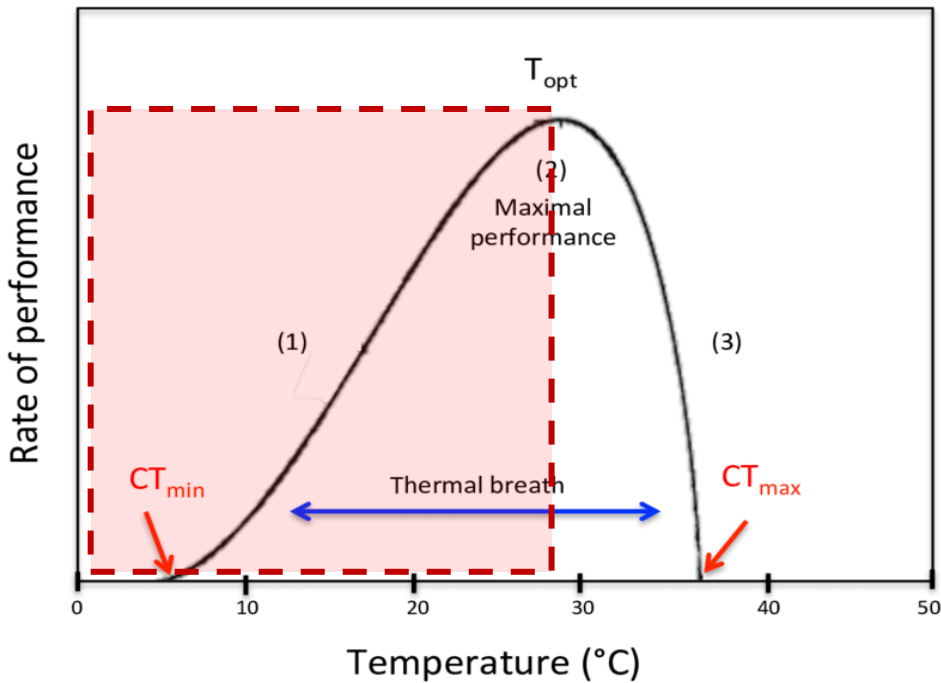
$$\ln(0.10) \approx -2.3026$$

$$\sqrt{-\ln(0.10)} = \sqrt{2.3026} \approx 1.5174$$

Solve for σ_p :

$$\sigma_p = \frac{18}{2 \times 1.5174} = \frac{18}{3.0348} \approx \mathbf{5.931}$$

Thermal Tolerance



Given target $T = 12^{\circ}\text{C}$ for Species A and $T_{opt} = 28^{\circ}\text{C}$

$$\sigma_p \approx \mathbf{5.931}$$

Fitting Thermal Performance Curves, $P(T)$

Rise in performance up to the optimal temperature (T_{opt}) ($T \leq T_{opt}$)

$$P(T) = \exp\left\{-\left(\frac{T - T_{opt}}{2\sigma_p}\right)^2\right\}$$

$$T - T_{opt} = 12 - 28 = -16$$

$$2\sigma_p = 2 \times (5.931) = 11.862$$

$$\frac{-16}{11.862} \approx -1.348$$

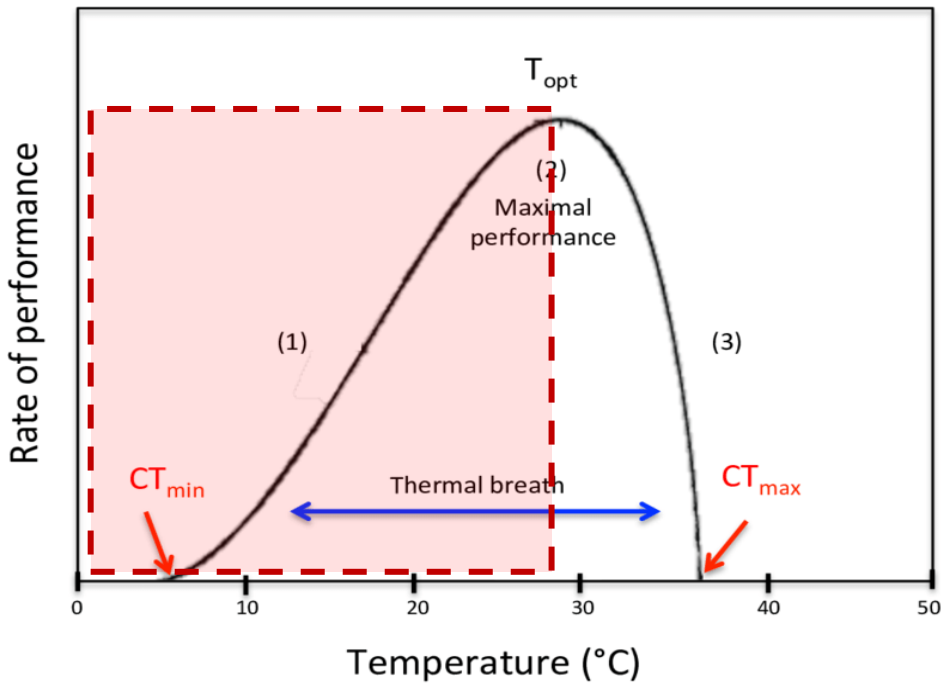
$$(-1.348)^2 \approx 1.819$$

$$\exp(-1.819) = e^{-1.819} \approx 0.162$$

$$0.162 \times 100\% \approx 16.22\% \quad (\approx 16.22\% \text{ in sample dataset})$$

- Therefore, at target $T = 12^{\circ}\text{C}$ and $T_{opt} = 28^{\circ}\text{C}$, the performance of species A is predicted to be 16.22%

Thermal Tolerance



Given target $T = 12^\circ\text{C}$ for Species A and $T_{\text{opt}} = 28^\circ\text{C}$

$$\sigma_p \approx 5.931$$

Fitting Thermal Performance Curves, $P(T)$

Rise in performance up to the optimal temperature (T_{opt}) ($T \leq T_{\text{opt}}$)

$$P(T) = \exp\left\{-\left(\frac{T - T_{\text{opt}}}{2\sigma_p}\right)^2\right\}$$

$$T - T_{\text{opt}} = 12 - 28 = -16$$

$$2\sigma_p = 2 \times (5.931) = 11.862$$

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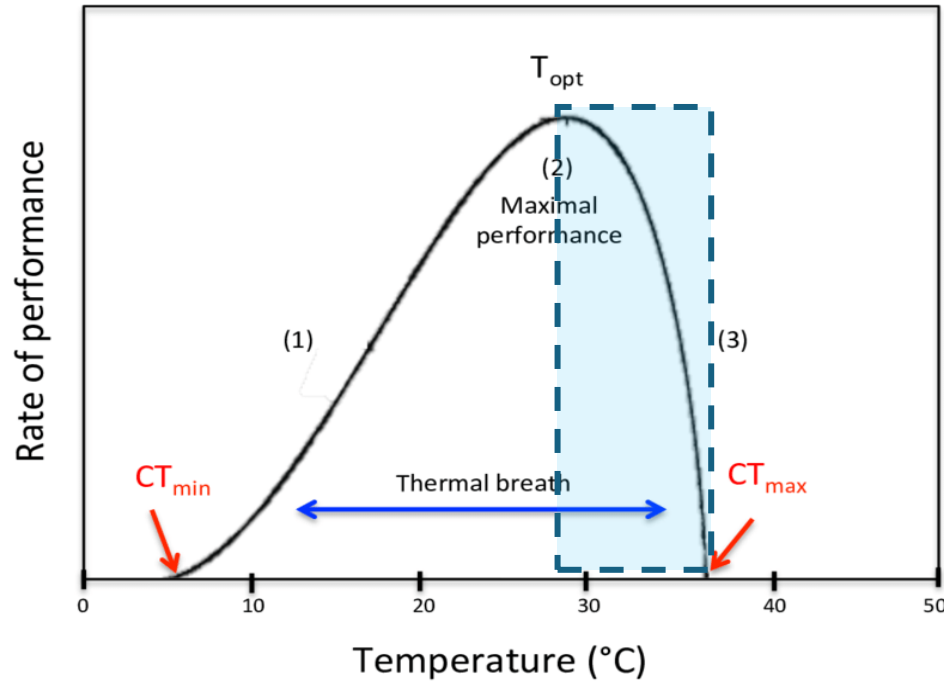
- Once you have established or fitted the values for T_{opt} and the performance width parameter σ_p for a given species, the cold-side piecewise formula acts as a **predictive model**.
- You can plug in **any arbitrary temperature** (including temperatures you never experimentally tested) to predict performance as long as target $T \leq T_{\text{opt}}$

Thermal Tolerance

Fitting Thermal Performance Curves, $P(T)$

Above optimal temperature ($T > T_{\text{opt}}$)

$$P(T) = 1 - \left(\frac{T - T_{\text{opt}}}{T_{\text{opt}} - CT_{\text{max}}} \right)^2$$



- A inverted parabolic decay that models the rapid, steep decline in biological performance caused by heat stress and enzyme denaturation past T_{opt} .
- When $T = CT_{\text{max}}$, the fraction reduces to $1 - (1)^2 = 0$, meaning performance drops to zero. Beyond this point, thermal stress is fatal.

Thermal Tolerance

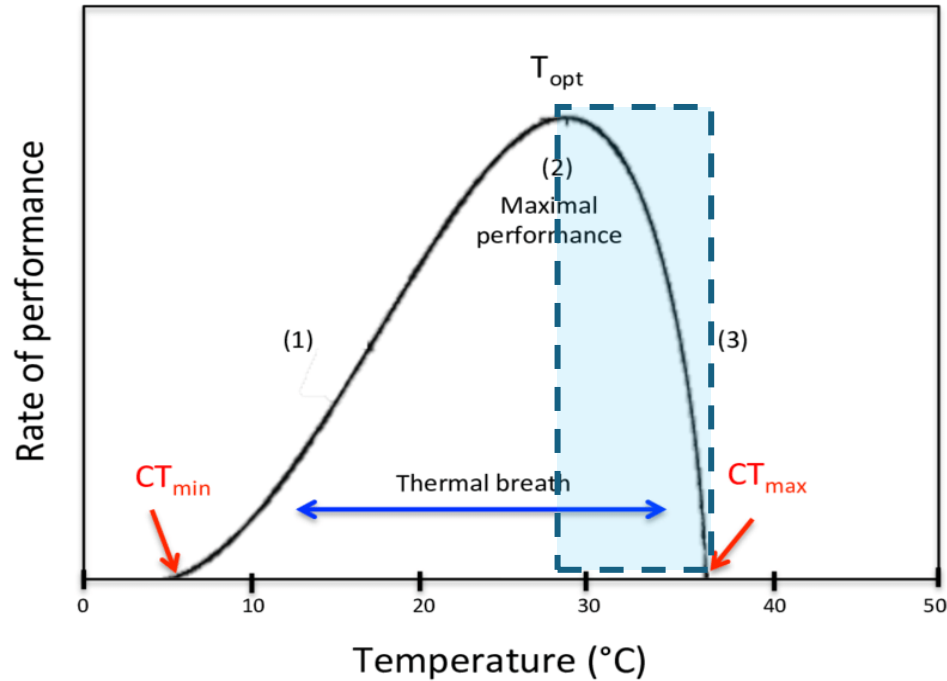
Fitting Thermal Performance Curves, $P(T)$

Above optimal temperature ($T > T_{\text{opt}}$)

$$P(T) = 1 - \left(\frac{T - T_{\text{opt}}}{T_{\text{opt}} - CT_{\text{max}}} \right)^2$$

Given Parameters:

- $T_{\text{opt}} = 28^{\circ}\text{C}$
- $CT_{\text{max}} = 36^{\circ}\text{C}$
- Target Temperature $T = 32^{\circ}\text{C}$



$$T - T_{\text{opt}} = 32 - 28 = 4$$

$$T_{\text{opt}} - CT_{\text{max}} = 28 - 36 = -8$$

$$\frac{4}{-8} = -0.5$$

$$(-0.5)^2 = 0.25$$

$$P(32) = 1 - 0.25 = 0.75$$

$$0.75 \times 100\% = 75\%$$

Thermal Tolerance

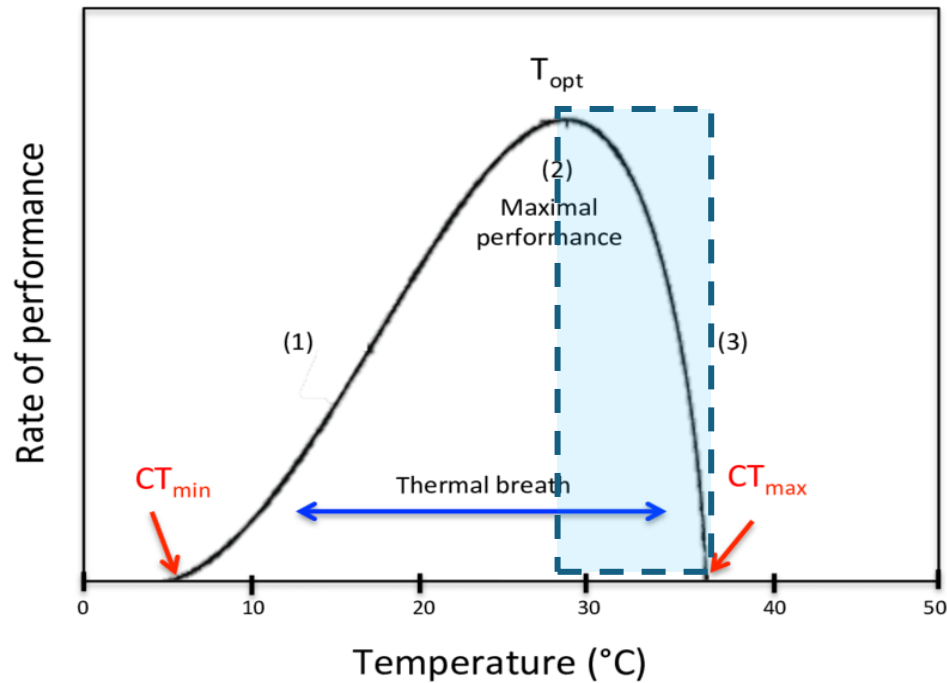
Fitting Thermal Performance Curves, $P(T)$

Above optimal temperature ($T > T_{\text{opt}}$)

$$P(T) = 1 - \left(\frac{T - T_{\text{opt}}}{T_{\text{opt}} - CT_{\text{max}}} \right)^2$$

Given Parameters:

- $T_{\text{opt}} = 28^{\circ}\text{C}$
- $CT_{\text{max}} = 36^{\circ}\text{C}$
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$$T - T_{\text{opt}} = 32 - 28 = 4$$

$$T_{\text{opt}} - CT_{\text{max}} = 28 - 36 = -8$$

$$\frac{4}{-8} = -0.5$$

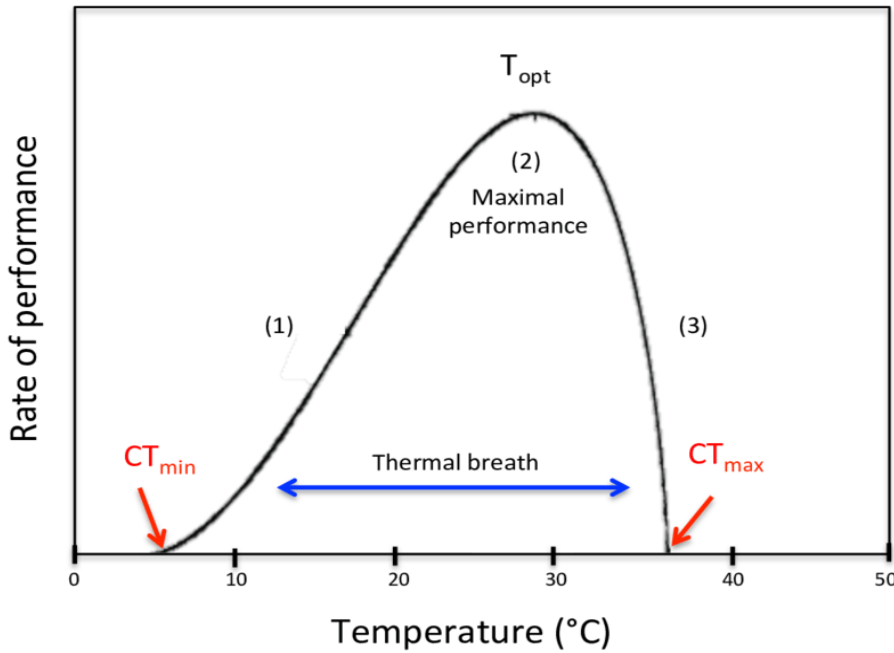
$$(-0.5)^2 = 0.25$$

$$P(32) = 1 - 0.25 = 0.75$$

$$0.75 \times 100\% = 75\%$$

- Unlike the cold side (which fit the 12°C data point very closely), the standard parabolic hot-side formula predicts a more gradual decline (75%) than what Species A actually experienced (16.22%).
- This tells us that Species A suffers a much sharper, non-linear collapse in fitness once it exceeds its optimal temperature, a common trait in temperature-sensitive organisms where heat stress rapidly disrupts physiological processes.

Thermal Tolerance



Fitting Thermal Performance Curves, $P(T)$

Zero OR non-zero Critical Thermal Limits?

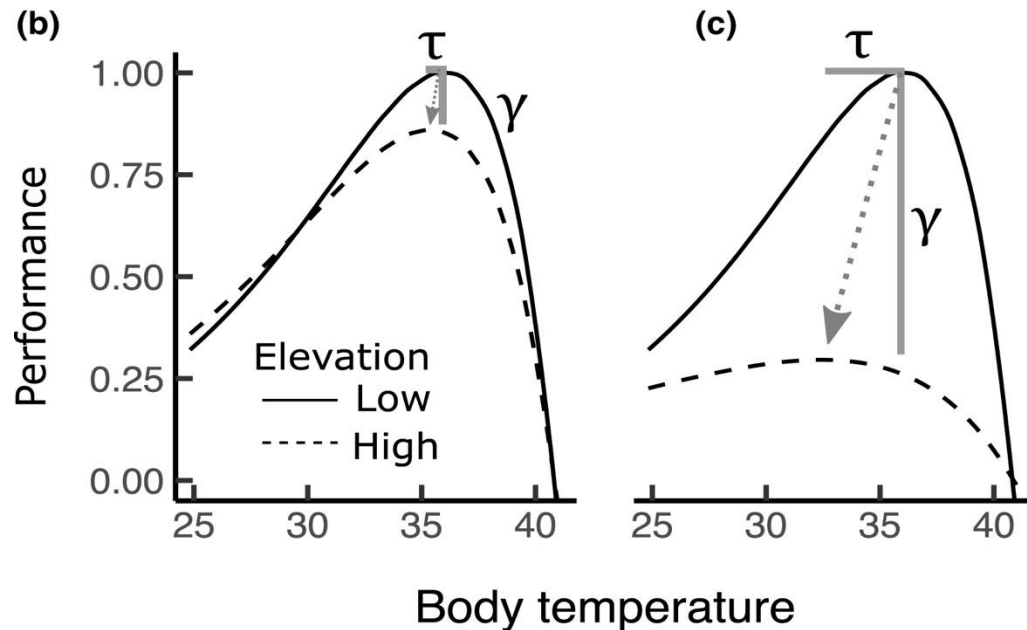
$$P_{CT_{min}} = 0 \vee P_{CT_{min}} \neq 0$$

$$P_{CT_{max}} = 0 \vee P_{CT_{max}} \neq 0$$



Comes down to three main factors:

- the trait being measured
- experimental constraints, and
- mathematical modeling choices



Thermal Tolerance

Fitting Thermal Performance Curves, $P(T)$

Zero OR non-zero Critical Thermal Limits?

$$P_{CT_{min}} = 0 \vee P_{CT_{min}} \neq 0$$
$$P_{CT_{max}} = 0 \vee P_{CT_{max}} \neq 0$$


1. Type of Trait Measured (Physiological Reality)

- **Locomotor & Coordinated Traits (Can Reach Zero):** Traits like sprint speed, swimming velocity, feeding rate, and escape response depend on neuromuscular coordination. As temperatures approach CT_{min} or CT_{max} , ion channels fail, muscle contraction ceases, or loss of righting response occurs. Because organisms become completely incapacitated, performance drops to **exactly zero**.
- **Basal & Metabolic Traits (Do Not Reach Zero):** Traits like resting metabolic rate, heart rate, or oxygen consumption do not hit zero at thermal limits. Standard metabolism often increases exponentially with temperature right up until thermal collapse or tissue death occurs. Because basal cellular respiration continues until the organism dies, these curves end abruptly above zero rather than tapering down to zero.

Thermal Tolerance

Fitting Thermal Performance Curves, $P(T)$

Zero OR non-zero Critical Thermal Limits?

$$P_{CT_{\min}} = 0 \vee P_{CT_{\min}} \neq 0$$
$$P_{CT_{\max}} = 0 \vee P_{CT_{\max}} \neq 0$$

?

2. Experimental Limits & Data Truncation

- **Avoiding Animal Mortality:** Researchers frequently stop measurements before reaching true $CT_{\min}$ or $CT_{\max}$ to avoid killing or causing irreversible heat/cold damage to experimental animals.
- **Measuring Non-Lethal Ranges:** When data are collected only within a sub-lethal range (e.g., 10°C to 35°C), the plotted curve ends at the extreme data points measured, leaving non-zero values at both ends.

Thermal Tolerance

Fitting Thermal Performance Curves, $P(T)$

Zero OR non-zero Critical Thermal Limits?

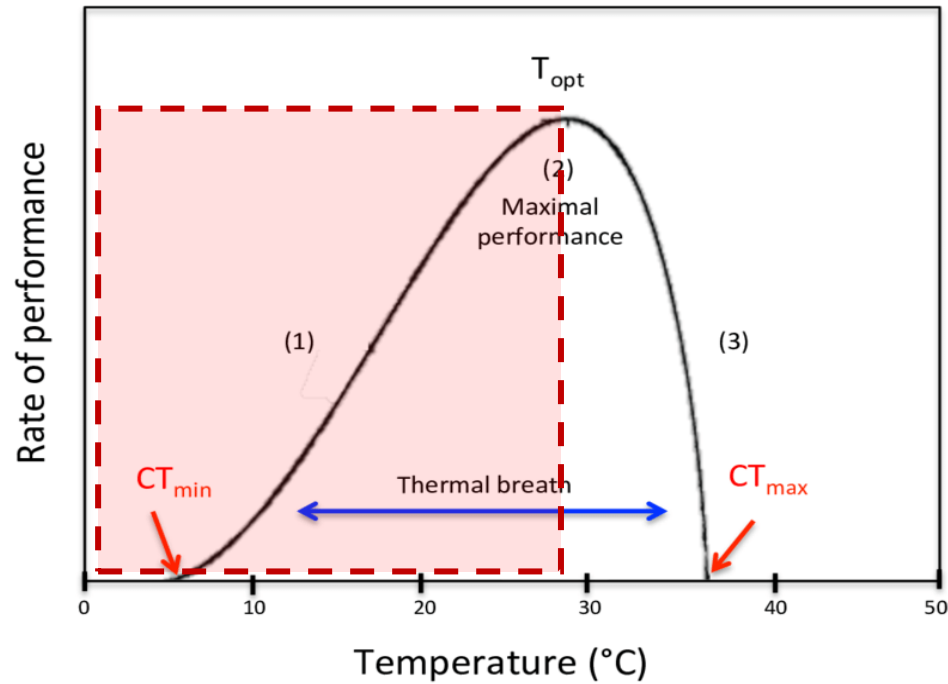
$$P_{CT_{min}} = 0 \vee P_{CT_{min}} \neq 0$$
$$P_{CT_{max}} = 0 \vee P_{CT_{max}} \neq 0$$


3. Mathematical Models & Curve Fitting

- **Bounded Models:** Mathematical formulations specifically built for biological performance (such as modified Sharpe-Schoolfield or Briere models) explicitly define performance as zero outside the lower and upper critical bounds $[CT_{min}, CT_{max}]$.
- **Unbounded/Empirical Fits:** Standard polynomial or Gaussian regression models simply fit a smooth line to experimental data points. These mathematical equations may yield non-zero values at the empirical endpoints or asymptote toward zero without ever touching it.

Thermal Tolerance

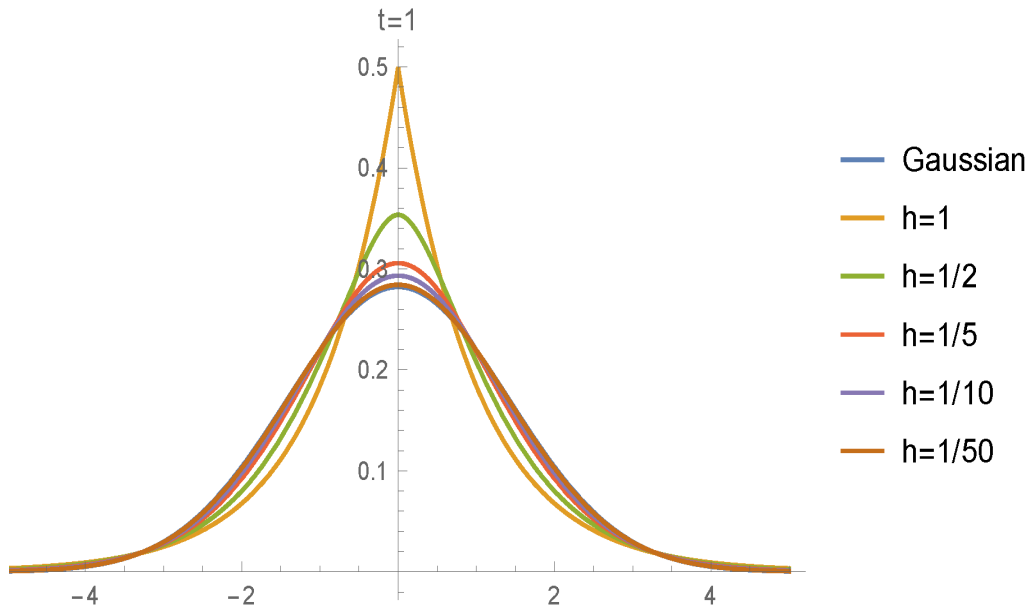
Fitting Thermal Performance Curves, $P(T)$



- Rise in performance up to the optimal temperature (T_{opt}) ($T \leq T_{opt}$)
- Uses a Gaussian function

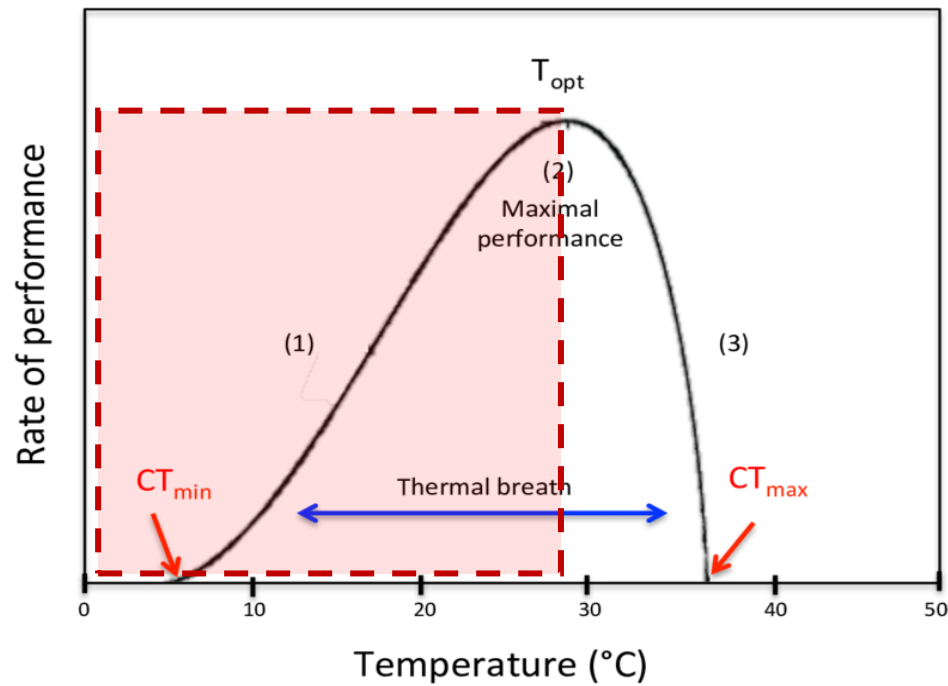
$$P(T) = \exp \left\{ - \left(\frac{T - T_{opt}}{2\sigma_p} \right)^2 \right\}$$

- A standard Gaussian function never touches zero
- Asymptotic to the x-axis



$$\lim_{x \rightarrow \infty} f(x) \rightarrow 0 \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) \rightarrow 0$$

Thermal Tolerance



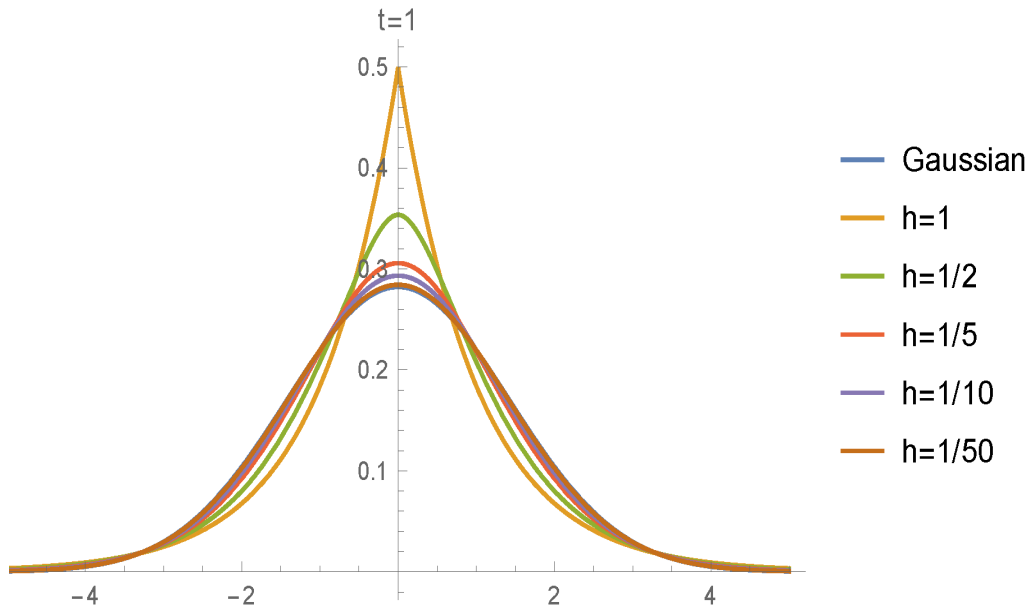
Fitting Thermal Performance Curves, $P(T)$

- Rise in performance up to the optimal temperature (T_{opt}) ($T \leq T_{opt}$)
- Therefore, for a Gaussian performance...

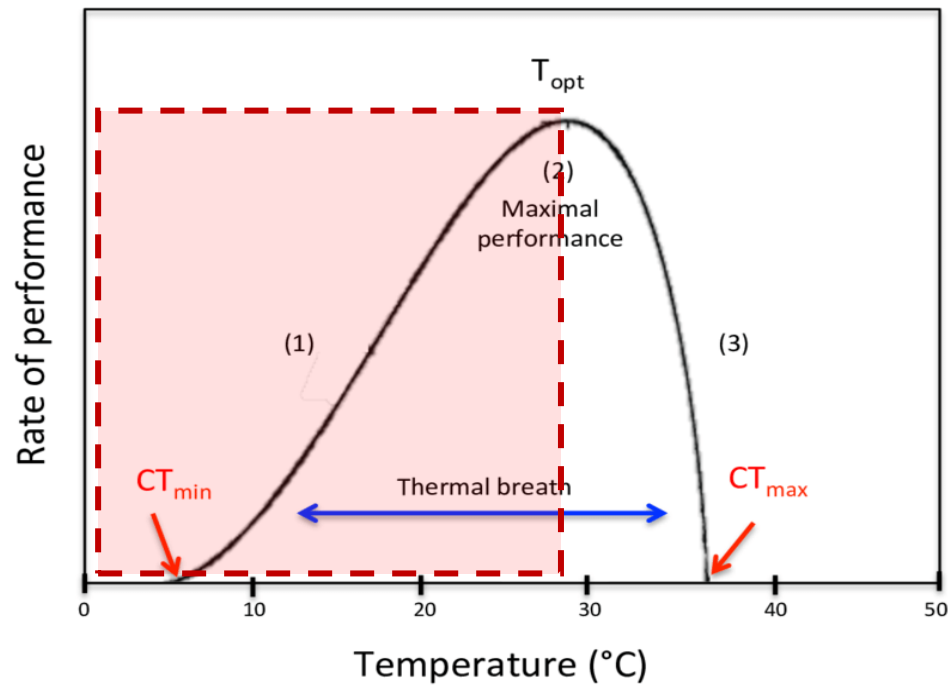
$$P(T) = \exp \left\{ - \left(\frac{T - T_{opt}}{2\sigma_p} \right)^2 \right\} > 0 \quad \text{for all } T$$

However...

- In reality, an organism's biological performance does reach an absolute zero point (CT_{min}) where the organism loses function completely.
- Because CT_{min} does not naturally fall out as a zero point in this Math, it cannot enter directly into the basic Gaussian function.



Thermal Tolerance



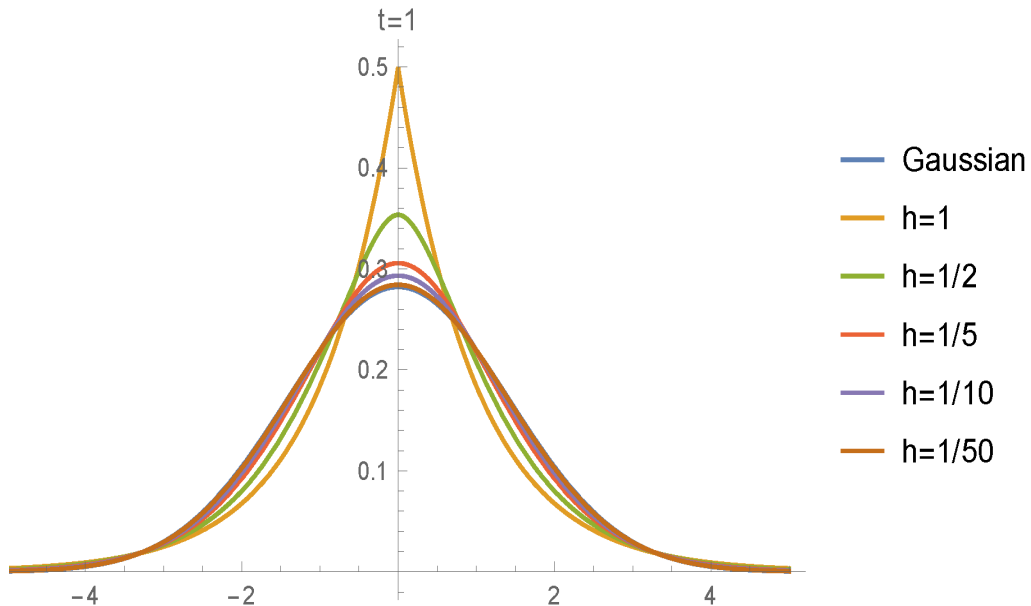
Fitting Thermal Performance Curves, $P(T)$

- Rise in performance up to the optimal temperature (T_{opt}) ($T \leq T_{opt}$)
- Therefore, for a Gaussian performance...

$$P(T) = \exp \left\{ - \left(\frac{T - T_{opt}}{2\sigma_p} \right)^2 \right\} > 0 \quad \text{for all } T$$

However...

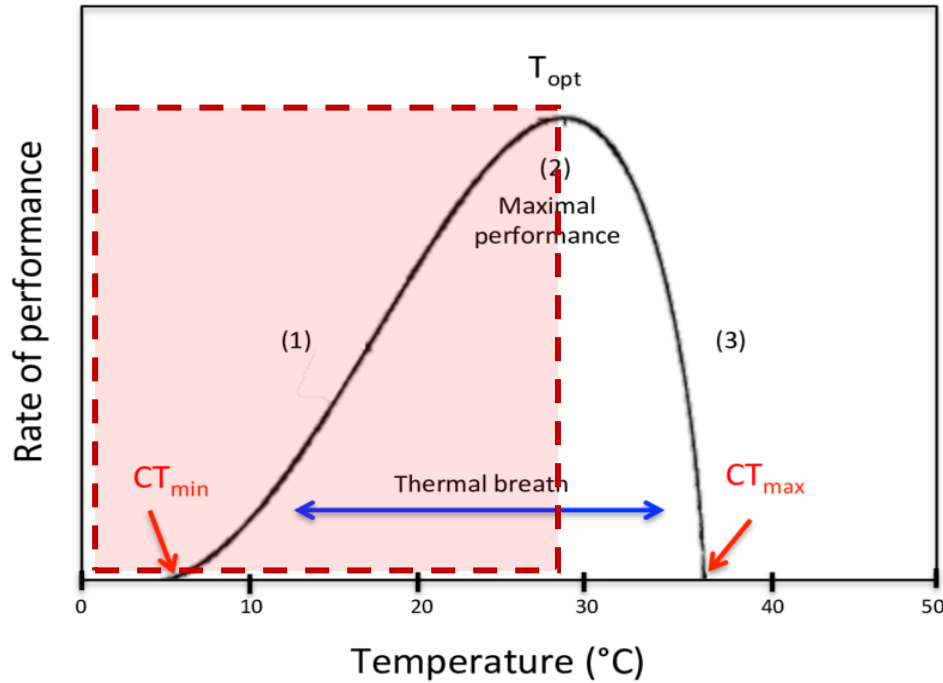
- In reality, an organism's performance is not a Gaussian function. Each an absolute zero performance temperature (CT_{min}) and a maximum performance temperature (CT_{max}).
- Because of this, the Gaussian function is not a good fit for thermal performance curves. Math, however, can be used to fit the data into the basic Gaussian function.



So, How is CT_{min} Treated in the Gaussian Function?

Thermal Tolerance

Fitting Thermal Performance Curves, $P(T)$



- Rise in performance up to the optimal temperature (T_{opt}) ($T \leq T_{opt}$)
- Therefore, for a Gaussian performance...

$$P(T) = \exp\left\{-\left(\frac{T - T_{opt}}{2\sigma_p}\right)^2\right\} > 0 \quad \text{for all } T$$

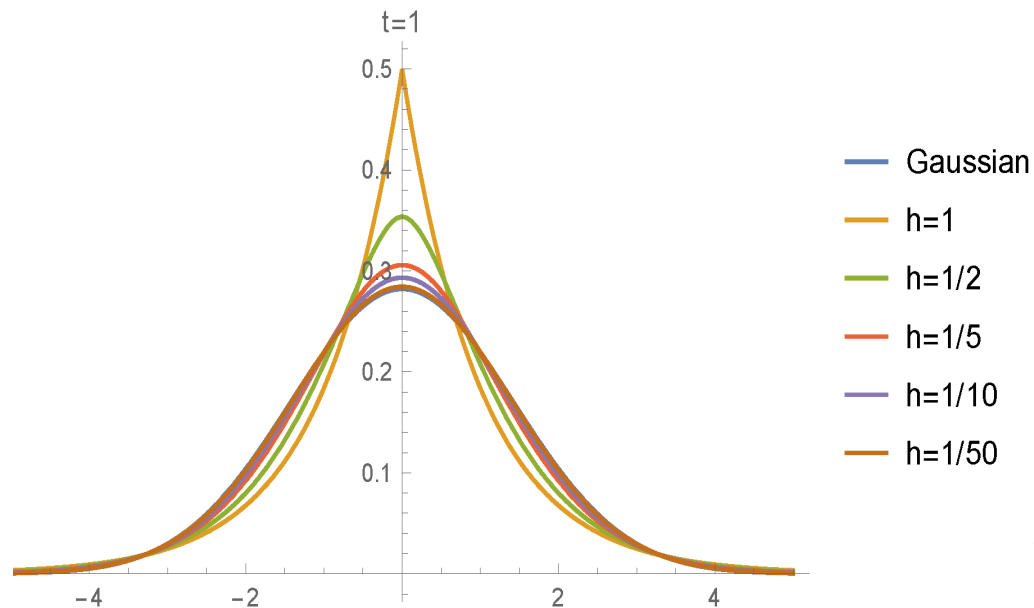
- Set a specific, fixed threshold for what constitutes "near-zero performance"
- We can define an **operational** CT_{min} fixed precisely at $4\sigma_p$ below T_{opt} :

$$CT_{min} = T_{opt} - 4\sigma_p$$

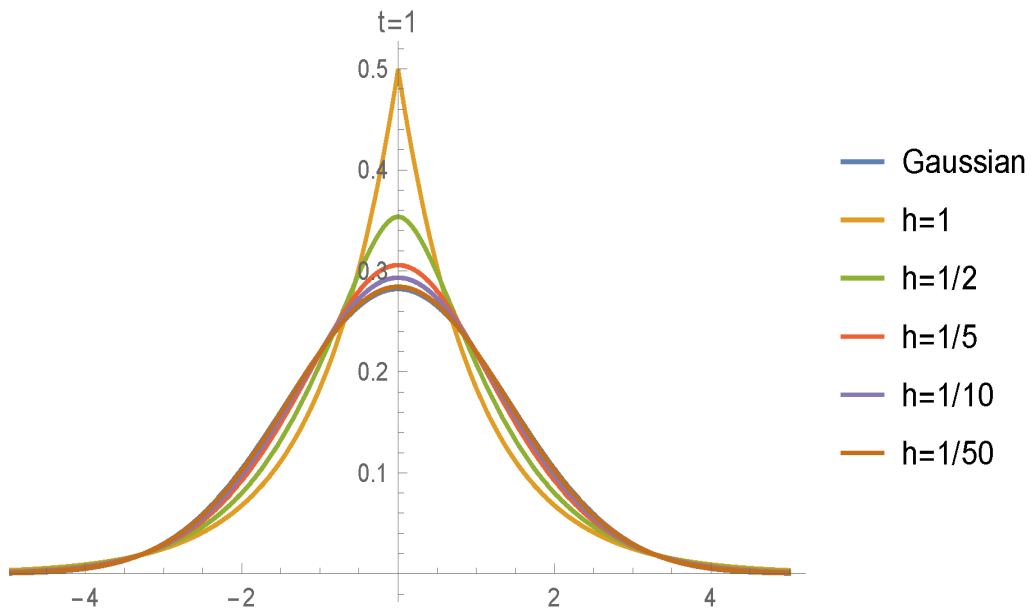
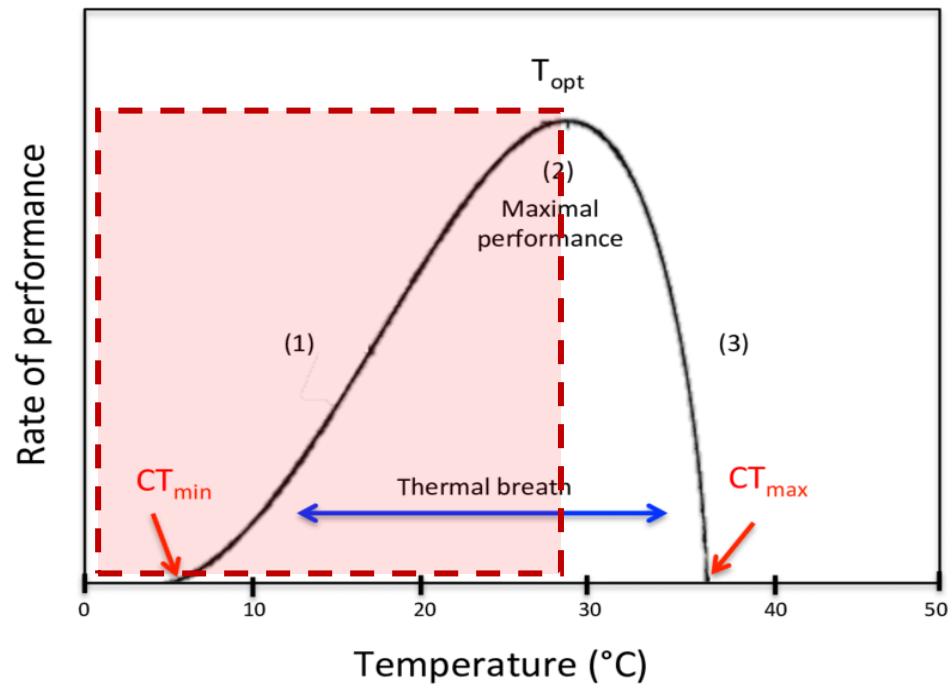
- Plug $T = T_{opt} - 4\sigma_p$ into the Gaussian cold-side equation:

$$P(CT_{min}) = \exp\left\{-\left(\frac{-4\sigma_p}{2\sigma_p}\right)^2\right\} = \exp(-2^2) = e^{-4} \approx 0.0183 \quad (\approx \mathbf{1.8\%})$$

Therefore, instead of forcing the curve to hit zero, we define CT_{min} as the temperature where performance drops to a negligible floor of 1.8% of peak performance.



Thermal Tolerance



Fitting Thermal Performance Curves, $P(T)$

- Since,

$$P(CT_{min}) = \exp\left\{-\left(\frac{-4\sigma_p}{2\sigma_p}\right)^2\right\} = \exp(-2^2) = e^{-4} \approx 0.0183 \quad (\approx \mathbf{1.8\%})$$

- We can then define the actual value for **Operational CT_{min}** where the negligible performance is 0.0183 and be able to plot CT_{min} on the curve while using the Gaussian function which would, otherwise, not accommodate for $P(CT_{min}) = 0$ under normal circumstances.
- Based on earlier derivation on **Slide 11**, we find **Operational CT_{min}** as T :

$$\sqrt{-\ln(P(T))} = \frac{|T - T_{opt}|}{2\sigma_p}$$

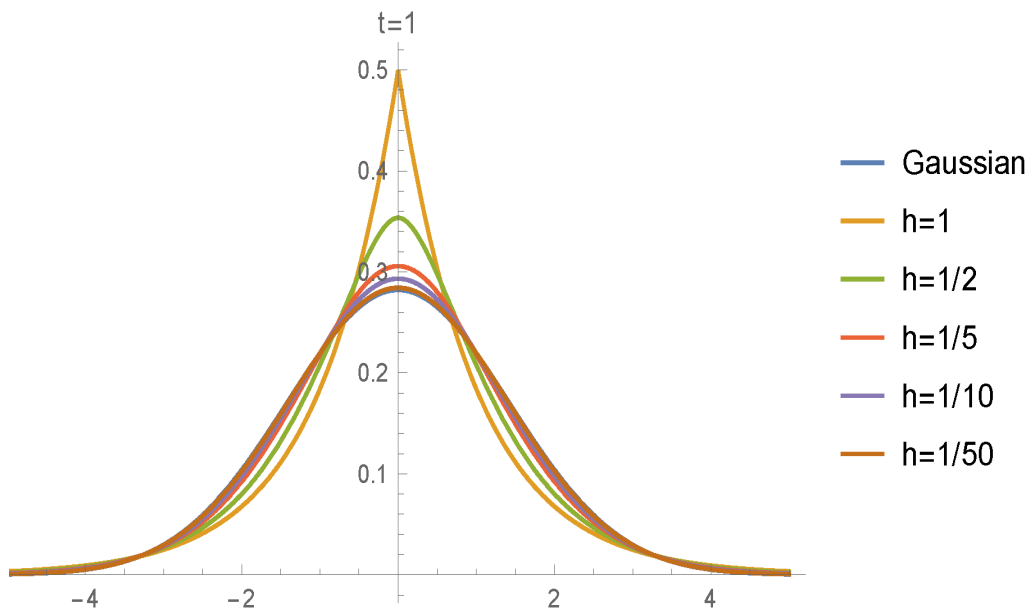
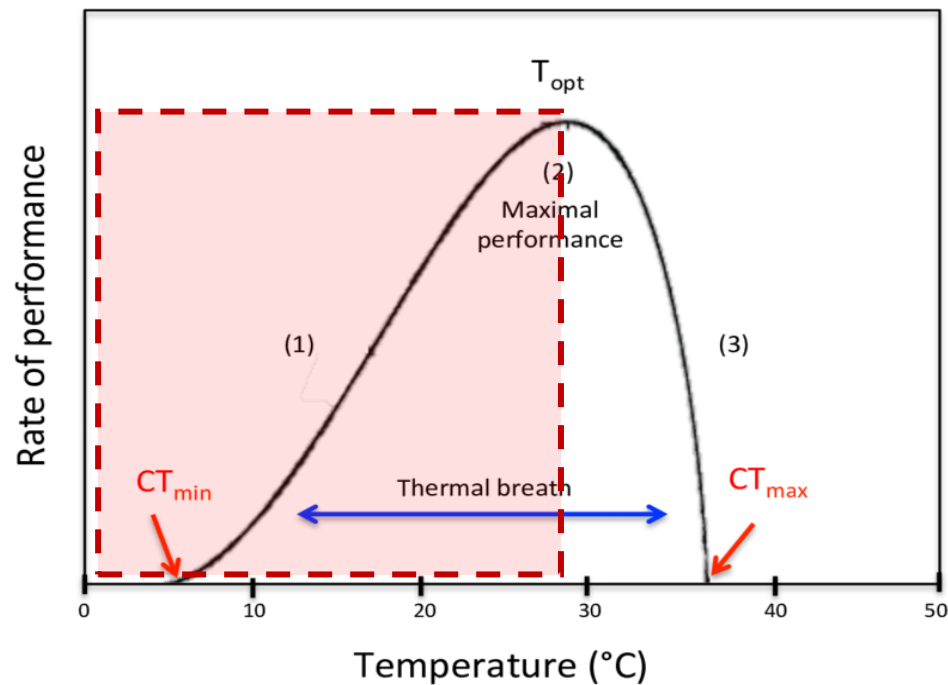
Since $T \leq T_{opt}$, we can write $|T - T_{opt}|$ explicitly as $(T_{opt} - T)$:

$$\sqrt{-\ln(P(T))} = \frac{T_{opt} - T}{2\sigma_p}$$

$$2\sigma_p \sqrt{-\ln(P(T))} = T_{opt} - T \quad \longrightarrow \quad T = T_{opt} - 2\sigma_p \sqrt{-\ln(P(T))}$$

Continued on the next slide...

Thermal Tolerance



Fitting Thermal Performance Curves, $P(T)$

$$P(CT_{min}) = \exp\left\{-\left(\frac{-4\sigma_p}{2\sigma_p}\right)^2\right\} = \exp(-2^2) = e^{-4} \approx 0.0183 \quad (\approx \mathbf{1.8\%})$$

$$T = T_{opt} - 2\sigma_p\sqrt{-\ln(P(T))}$$

$$T = T_{opt} - 2\sigma_p\sqrt{-\ln(0.0183)}$$

$$T = T_{opt} - 2\sigma_p\sqrt{-(-4)}$$

$$T = T_{opt} - 2\sigma_p(2)$$

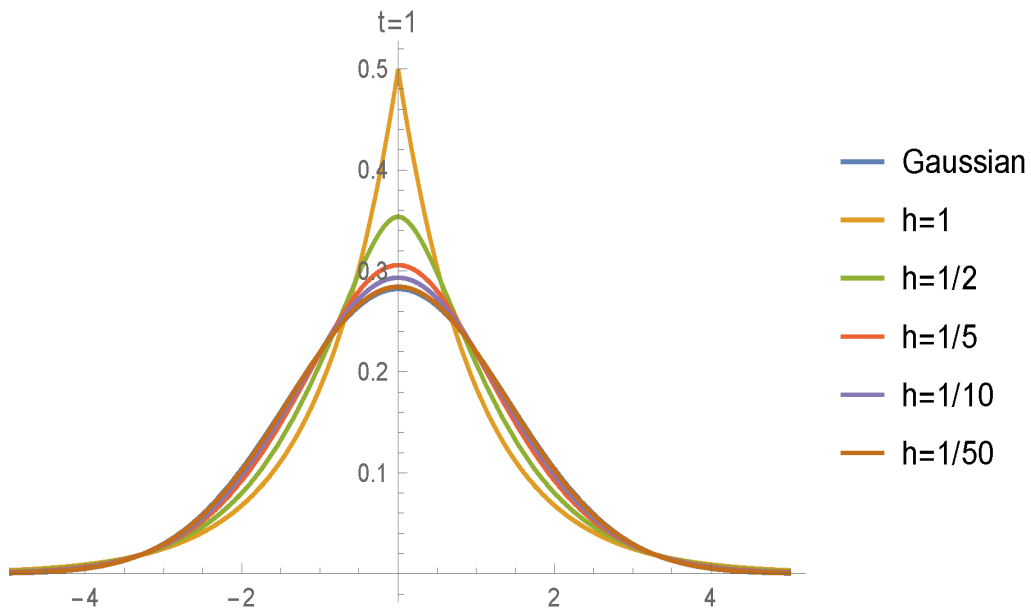
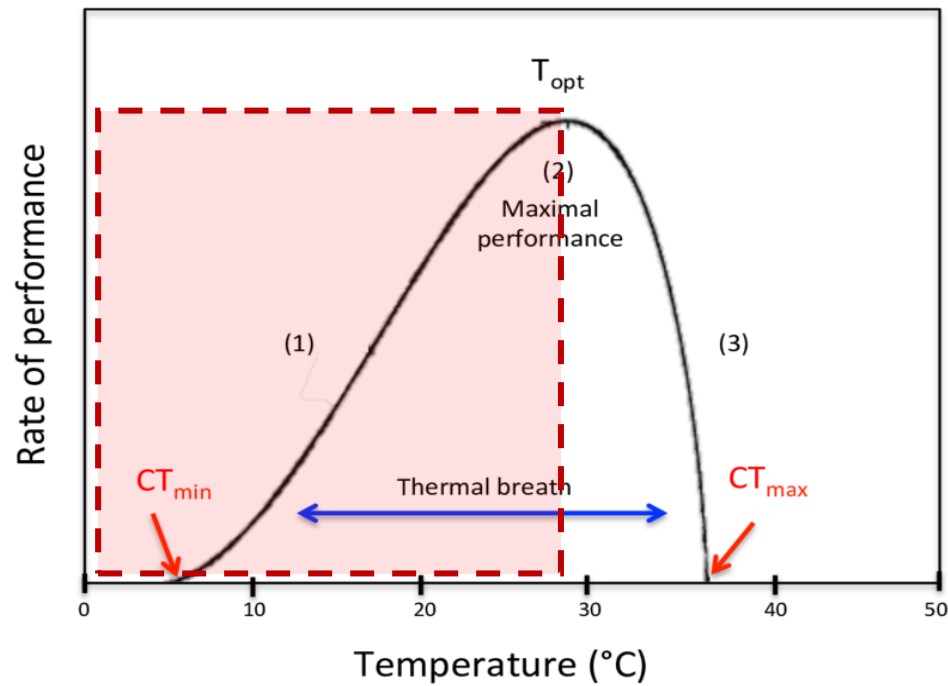
$$T = CT_{min} = T_{opt} - 4\sigma_p$$

Supposed $T_{opt} = 28^\circ\text{C}$, $\sigma_p \approx 5.931$ and $P(CT_{min}) = 0.0183$

$$CT_{min} = 28 - 4(5.931) = 4.276$$

Therefore, at $P(CT_{min}) = 0.0183$, $CT_{min} = 4.276$, we can plot this!

Thermal Tolerance



Fitting Thermal Performance Curves, $P(T)$

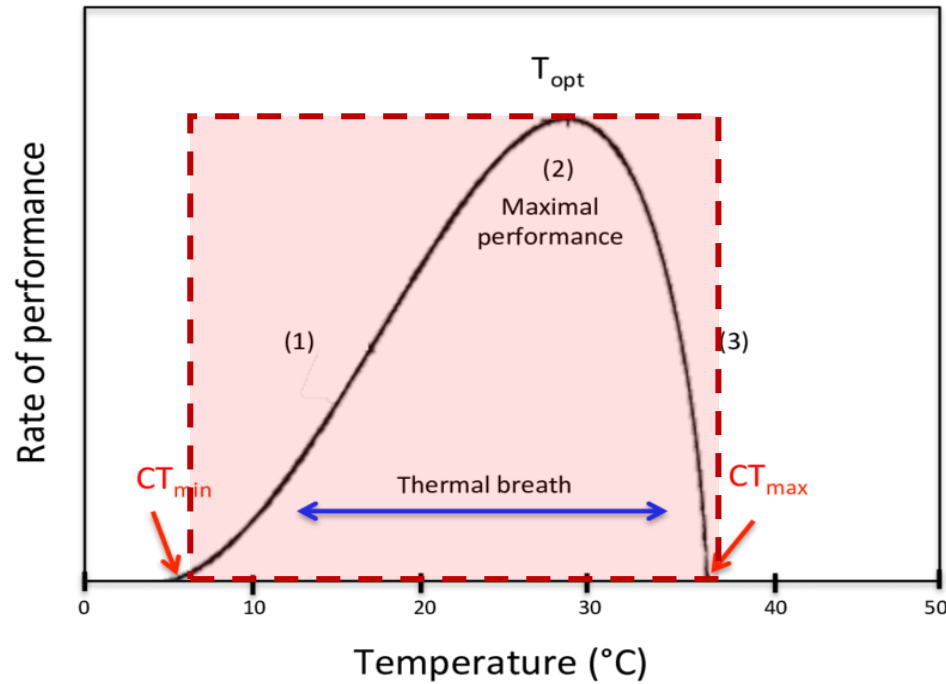
- When you evaluate the Gaussian cold-side equation at a temperature T that is k width units (σ_p) below T_{opt} (meaning $T = T_{opt} - k\sigma_p$):

$$P(T) = \exp\left\{-\left(\frac{-k\sigma_p}{2\sigma_p}\right)^2\right\} = \exp\left\{-\left(\frac{-k}{2}\right)^2\right\} = e^{-(k/2)^2}$$

Notice how σ_p cancels out completely. The predicted performance percentage depends **only** on the multiplier k :

- If $k = 1 \Rightarrow e^{-(1/2)^2} = e^{-0.25} \approx 77.9\%$ performance
- If $k = 2 \Rightarrow e^{-(2/2)^2} = e^{-1.00} \approx 36.8\%$ performance
- If $k = 3 \Rightarrow e^{-(3/2)^2} = e^{-2.25} \approx 10.5\%$ performance
- If $k = 4 \Rightarrow e^{-(4/2)^2} = e^{-4.00} \approx \mathbf{1.83\%}$ performance

Thermal Tolerance



Fitting Thermal Performance Curves, $P(T)$

Defining Curve Asymmetry (α)

- Defining a metric to measure how "lopsided" or asymmetric the curve is:

$$\alpha = \frac{\text{Warm Side Width}}{\text{Cold Side Width}} = \frac{CT_{max} - T_{opt}}{T_{opt} - CT_{min}}$$

Substituting the operational definition $CT_{min} = T_{opt} - 4\sigma_p$ into the denominator :

$$\alpha = \frac{CT_{max} - T_{opt}}{4\sigma_p}$$

What α Tells You:

- Ectotherm Physiology:** In almost all ectotherms, the hot side drops much faster than the cold side, meaning $CT_{max} - T_{opt}$ is smaller than $T_{opt} - CT_{min}$.
- Asymmetry Range:** This ratio α typically falls between 0.1 and 1.0, quantifying that organisms collapse rapidly under heat stress compared to cold stress.
- $\alpha = 1.0$ (**Symmetric**): The cold buffer and heat buffer are equal. If an animal tolerates 10°C below its peak, it can tolerate 10°C above it. (*Rarely happens in nature*).
- $\alpha = 0.2$ (**Typical Ectotherm**): The heat buffer is only 20% **as wide** as the cold buffer.



THANK YOU